WSN Infrastructure and Testing of Wireless Sensor Networks

Whitepaper

Abstract:

Why are WSN (Wireless Sensor Network) often an important part of an IoT system?

The Internet of Things (IoT), a network of connected smart devices providing the required data, is the solution for this request.

We have seen that one or more technologies are replaced by another technology in a short period of time. Earlier, we used analogue devices, and then progressed to centralized wiring systems, digital systems, and then extensive wireless systems.

Here comes the wireless sensing network (WSN), introduced as new concepts in our lives like smart cities, which will be used to achieve a technological goal that requires low-cost, low-power transceivers that are compact in size and have some standard protocol stacks. WSN is used across some of the main sectors, like health, agriculture, home, and office applications.

Keywords: WSN, Wireless Sensor Networks, IoT, real-time communication, low power consumption.

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Introduction

This paper focuses on the wireless sensor network (WSN) architecture, its applications, and its usage.

In the ecosystem, researchers, engineers, and innovators are looking closely at the Internet of Things (IoT) as a potential source of revenue.

Let us introduce some new methodologies for collecting data that will dramatically change how organizations learn about their surroundings and respond to changes, and there should be at least one new technology that is powerful and makes an impact.

Yes, we are discussing wireless sensor networks. They consist of a group of sensor nodes (which are interlinked using wireless communication) that are used to observe and monitor surrounding attributes like temperature, pressure, humidity, light, vibration, sound, etc.
To simulate or test WSN in an IoT system if we go with the prototype, we generally use Arduino, Raspberry Pi etc. For a prototype, this looks good, but when we go with real-world deployment, this is a different factor altogether.

In the real ecosystem, we see a variety of constraints: time constraints, network constraints, and cost constraints. Sometimes we need to make sure that solution works in hard environments.

As an example, let us consider a project regarding smart agriculture. This project might involve deployment of several hundreds or thousands of sensors all over a field covering an area of several meters to kilometers. Most often, these sensors must run on minimal batteries, and the radio network may have bandwidth and quality constraints. And the project might have some budget constraints. This is where the wireless sensor networks come into play.

A wireless sensor network consists of many sensor nodes arranged in a variety of topology configurations, like the star or mesh. These sensor nodes are usually powered by batteries or solar, and they are designed in such a way that they operate on very little power.

Depending on the implementation, sensor nodes will have a transmission range of 10 to 30 meters. Data from all sensors will be transmitted to the gateway, and then from the gateway to the cloud services via cellular or Wi-Fi networks.

Overview

The following are the topics to ensure coverage of WSN testing: The following sections go over the specifics of each topic.

I. WSN Architecture

II. WSN Infrastructure

III. WSN Testing

IV. WSN Vs MANET

V. WSN Applications

VI. Advantages of WSN

VII. Operational Challenges of WSN

I. WSN Architecture

We need to understand WSN components and their structures to know more details about WSN hardware. Wireless sensors in a network consist of very small microcontrollers, a sensor component, and a power supply. An example of WSN architecture is shown in Fig. 1. The wireless sensor nodes consist of a power supply, sensor component, and on-board processor that will enable local processing of some simple computations and send only the required processed data. Also, each node is provided with a radio transceiver as a communication component.

Sensor node: an autonomous sensor-equipped device. This is an autonomous device equipped with sensors.

Gateway or Data Collector: Gathers sensor data and acts as a gateway to other systems. This must be a data capture device, and it should also be connected to external systems to transmit sensor data via cellular network or Wi-Fi.

Dashboard (cloud service): stores sensor data and manages it, which should be the data storage management center.

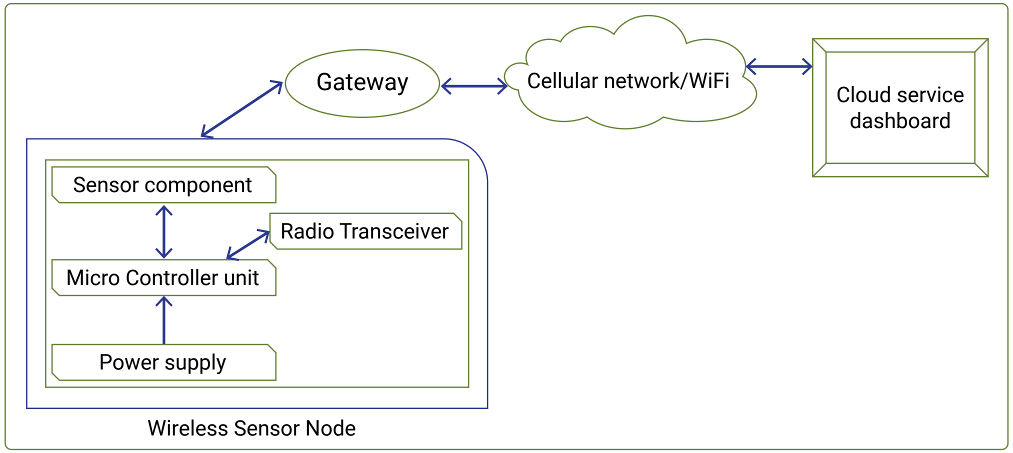

Fig. 1 WSN architecture

- Sensor
  An actuator that converts surrounding physical occurrences, e.g., temperature, vibration, motion, and sound, into electrical signals.
- Sensor node
  One of the most important and basic components in a sensor network is the board, which contains on-board sensors, a processor, memory, a transceiver, and a power supply.
- Sensor network
  Consists of many sensor nodes. Nodes can be deployed either inside or very close to the sensor environment.

II. **WSN Infrastructure**

How can we say that infrastructure is established? When the network is activated, the next step is to initiate a collaborative environment for the sensor network.

When the sensor network is activated, a variety of tasks must be performed to create and establish the necessary infrastructure that will allow useful collaborative work to be performed:

- Try to discover other nodes in the vicinity
- Adjusts its radio power to ensure proper connectivity
- It forms the cluster.
- Puts the node in a common temporal and special framework

Some common techniques used to establish the network are:

- Topology control
- Clustering
- Time synchronization
- Localization

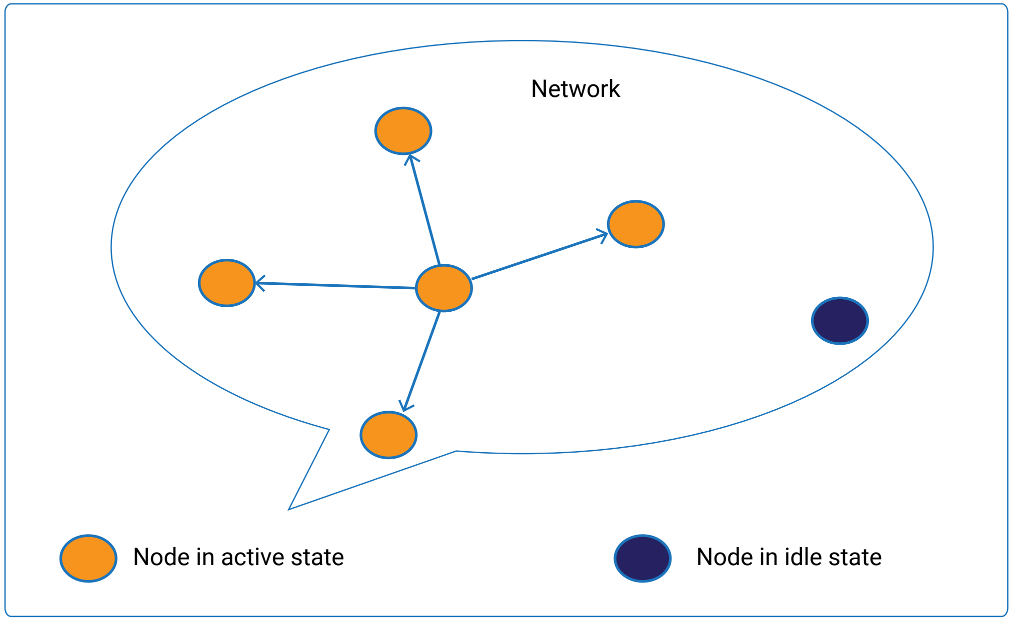

Fig. 2 Topology Control

1) Topology control

- It is one of the most important techniques used in wireless sensor networks to reduce energy consumption, which in turn is essential to extend the network's operational time and reduce radio interference.
- Basically, when a sensor node wakes up, it will try to discover other nodes in the vicinity with which it can communicate in a bidirectional way by executing a protocol.
- In the initial phase, each node in the network will try to connect with nearby nodes using its own radio link.
- To save energy and extend the network's lifetime, each sensor node will broadcast messages with less than its maximum possible radio power.

2) Clustering

In a network, architecture enables more efficient use of sensor resources, such as:

- Frequency spectrum
- Bandwidth
- Power

How the cluster is formed

- To differentiate each node, mechanism to assign unique ids (UIDs) to each node
- When the node has a higher ID than its uncovered neighbors, it declares itself a cluster head
- Nodes that fall under the same cluster head will communicate with each other
- A node that can communicate with two or more cluster heads may become a gateway

Gateway: Receives sensor data from different nodes and sends the data to the centralized system.

Uncovered neighbours: nodes that have not already been claimed by another cluster head.

3) Time Synchronization:

Time synchronization is the most crucial and important attribute in embedded systems. Each node runs independently, so their clocks may not be synchronized with each other. It is very important to run networks efficiently to detect events for localization by estimating internode distances.

4) Localization

One of the most extensive features in WSN is 'Localization,' which helps identify the current location of the sensor nodes. A wireless sensor network consists of hundreds of nodes, making the installation of GPS on each sensor node expensive. Another important factor is that GPS will not provide exact localization results in an indoor environment.

III. WSN Testing

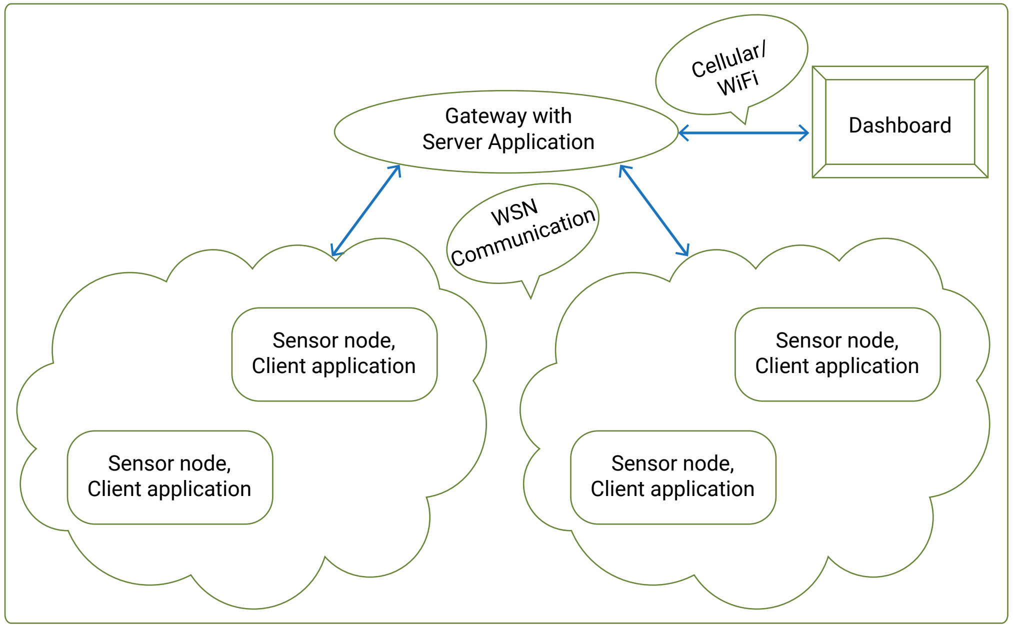

Fig. 3 WSN test environment

This section presents testing WSN software combines established methods from software engineering with WSN-specific tools, i.e., the execution of the software on a WSN platform in a real environment and the profiling of power consumption of sensor nodes.

Each sensor node has a unique ID to differentiate the component and will have client application software. As soon as the node wakes up and becomes active in the network, it tries to fetch the environmental sensor values (temperature, humidity, light, motion, vibration, etc.) and sends them to the gateway, which in turn posts the data to the cloud application. Gateway has a WSN server, which in turn will talk to the client application of the sensor node in the network. Once communication is established between the gateway and sensor node, messages will be exchanged. The Wireshark tool can be used to view these messages in over-the-air logs. The Wireshark tool collects pcap (packet capture) files that can be captured or recorded at each node and then opened in the Wireshark tool for the analysis of protocol messages. These messages will help to debug and test WSN in a real-world environment. There are other proprietary tools to capture WSN network logs and analyze the packet loss, connectivity issues with the server, node loss, etc.

IV. WSN Vs MANET

	Wireless Sensor Network	Mobile Adhoc Network
Standard	IEEE 802.15.4	IEEE 802.11
Communication mode	Broadcast	Point-to-point
Main purpose	Collect the information	Distributed computing
Interaction	More interaction with humans	More interaction with environment
Node movement	Centralized	Decentralized
Memory Constrain	Little high	Less than WSN

V. WSN Applications

Wireless sensor networks consist of different types of sensors, like temperature, ambient light, shock, tilt, pressure, low sampling rate, seismic, magnetic, thermal, visual, infrared, radar, and acoustic, which are cleverly used to monitor a wide range of ambient situations. These sensor nodes are used for constant sensing of values from surroundings, event ID, event detection, and local control of actuators. The applications of wireless sensor networks include health, agriculture, home, logistics, military, and other commercial areas.

- **WSN in Healthcare:**

In the healthcare sector, WSN uses advanced medical sensors to monitor and track patients within a healthcare facility and to provide real-time monitoring of patients by utilizing wearable hardware.

These days, most of us use smartphones. It essentially provides personal medicine care assistance by utilizing real-time sensors embedded in smartphones as well as a barcode system. The mechanism developed performs ECG monitoring in real time. Also, the monitoring of the blood sugar level, blood pressure, and several kinds of other diagnostics could be possible by using real-time sensors.

- **WSN in Agriculture:**

We see that agriculture is the largest system in the world that is not yet completely digitalized. It is a biological production system with many factors affecting its complexity, for instance, human behavior, machines, nature, chemicals, biology, weather, and climate.

Within the agriculture domain, the preservation of the crops plays a vital role. To provide a better environment for the crops, various WSN applications can be implemented. Crop irrigation and fertilization are some examples of these applications.

- **WSN in Home Automation**

Have we become a little lazy or too smart? Most of us have a smart phone and a smart home. When it comes to a smart home, network configuration and setup are the critical components for home security and automation. The devices (nodes) and gateway placed in the home will communicate with each other. One thing to note is that the application of a wireless sensor network in an IoT smart home is interesting. When can we say that it's home automation? It is at this point that the automation of activities begins. For example: control of lights, appliances, AC, heating, and door and gate safety locks.

- **WSN in transport logistics:**

Many logistics systems need real-time monitoring of various environmental parameters and better handling of packages. To track, trace, and monitor the shipment, transport logistics require WSNs to be used.

We are seeing an increase in the number of logistics shipments that are lost, damaged, or do not arrive on time. How do we track them? WSNs can be used in many logistics systems that need real-time monitoring of various environmental parameters and better handling of packages. These requirements can be fulfilled by combining the logistics systems with WSNs.

The transport logistics sector requires low costs and high-quality during deliveries. The development and deployment of a WSN-based system for monitoring transportation conditions such as temperature and humidity within a cargo container travelling via road, ship, or air freight improves quality by allowing for more supervision and lowers costs by reducing shipment loss or damage during transportation.

VI. **Advantages of WSN**

- Helps to avoid a lot of manual efforts. One example: checking the water level at the reservoir by deploying sensor nodes at the site and sending the information to the center system.
- WSN helps enable long-distance data collection and transmission
- WSN will help anticipate natural disasters
- WSN helps to avoid a lot of wiring
- WSN can be accessed and monitored through a centralized system

VII. **Operational Challenges of WSN**

- One major challenge is limited storage
- We see that WSN has low bandwidth and has high error rates
- WSN errors are common due to
    - » Wireless communication
    - » Value measurement in a noisy environment
    - » Node failures that are expected
- Data packets loss in noisy environments
- Scalability for many sensor nodes

Conclusion

WSN is possible and exists today due to tremendous technological advancement in various domains, and it has also become an essential part of our daily lives. System testing of WSNs is a little complex. This paper contributes to the WSN architecture, applications, and one of the methods for testing WSNs. One of the goals is to have an automated tool for executing test cases, extracting meaningful information from test executions, analyzing the monitored information, and checking for expected behavior.

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About the Author

Basavaraj Patil is a Principal Engineer with Sasken Technologies Ltd., having 14 years of experience in automation and testing projects across different areas like connectivity protocols (NFC, Bluetooth), the Internet of Things, Android, and multimedia.

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